

KneeSchool.com

Master Operations Handbook

A complete playbook for building the knee education platform, content engine, source repository, AI workflow, quality control system and commercial integration model.

Prepared for: Project assistant, content lead, medical writer, editor, project manager and clinical reviewer

Contents

This handbook is structured as a sequential operational manual. Use the heading navigation pane in Word for rapid movement through the document. A dynamic table of contents can be inserted or updated in Word before final publication.

1. Executive Summary

Purpose

KneeSchool.com is intended to become the definitive digital education platform for anyone who wants to understand the knee. The platform is not a clinic brochure. It is an encyclopaedia, learning pathway, case library, video academy, examination resource and research hub. It should serve patients, undergraduate medical students, postgraduate trainees, FRCS candidates, fellows and advanced knee consultants.

Strategic Role

The platform should educate first, build trust second, and support commercial conversion third. It will create authority for the founder, generate appropriate clinical referrals to Sportshealing, and provide contextual product pathways to OmKneeHealth.

Operating Model

The content engine will use curated source material, AI-assisted drafting, assistant-led organisation, evidence verification and consultant-level quality control. The assistant does not need to be a knee expert, but must follow strict source, prompt, editorial and verification procedures.

Success Definition

Success means creating a large, accurate, searchable and trusted knee knowledge platform that users return to repeatedly. It should feel closer to a specialist university and modern learning platform than a conventional healthcare website.

2. Vision, Mission and Strategic Positioning

Vision

To build the world's most useful and authoritative online knee education ecosystem, from basic patient education to advanced fellowship and consultant-level knee knowledge.

Mission

To make knee knowledge clear, structured, evidence-based and accessible to every learner level while preserving clinical accuracy and academic credibility.

Positioning Statement

KneeSchool is the complete knee knowledge hub: an educational platform for patients, students, trainees, fellows, consultants, therapists and researchers.

Brand Promise

Learn the knee properly. Understand the science. Apply the knowledge safely.

Tone

Expert, concise, clear, educational, evidence-led, visually engaging and clinically responsible.

Not the Brand

Not a generic orthopaedic blog. Not a supplement sales site. Not a hospital marketing site. Not an unstructured archive of articles. Not a superficial patient leaflet library.

3. Complete Sitemap

Top-Level Navigation

Home; Learn by Level; Knee Encyclopaedia; Conditions; Surgery Academy; Rehabilitation Academy; Research Academy; Case Library; Question Bank; Video Library; Knee Score Centre; Products and Services; Community; About.

Home Page Modules

Hero search bar; choose your learning path; interactive knee model; topic cards; featured modules; latest research; case challenge; video spotlight; Sportshealing referral prompt; OmKneeHealth product prompt; newsletter capture.

Learn by Level

Patient Academy; Medical Student Academy; MRCS Academy; FRCS Orth Academy; Fellowship Academy; Consultant Masterclass.

Knee Encyclopaedia

Anatomy; Biomechanics; Clinical Examination; Imaging; Conditions; Surgery; Rehabilitation; Performance; Women's Knee Health; Children's Knee Health; Research; Outcomes; Glossary.

Commercial Sections

Sportshealing Clinic pathway; OmKneeHealth product pathway; online courses; webinars; conference links; educational downloads.

Expanded Sitemap Table

Area	Included Modules
Home	Hero search, learning paths, interactive knee, featured content, latest research, case challenge, CTA
Learn by Level	Patient, Medical Student, MRCS, FRCS, Fellowship, Consultant
Encyclopaedia	Anatomy, biomechanics, examination, imaging, conditions, surgery, rehabilitation
Research	Methods, statistics, critical appraisal, landmark papers, controversies
Commercial	Sportshealing, OmKneeHealth, courses, webinars, events

4. Curriculum by Educational Level

Patient Academy

Clear explanation of knee anatomy, common symptoms, diagnosis, imaging, treatment options, rehabilitation, prevention and when to seek specialist help.

Medical Student Academy

Core anatomy, biomechanics, pathology, history, examination, imaging basics and common diagnoses.

MRCS Academy

Applied anatomy, examination technique, trauma principles, imaging interpretation, common pathology, viva preparation and SBA question practice.

FRCS Orth Academy

Advanced anatomy, surgical decision-making, evidence appraisal, complex cases, revision concepts, complications and viva-style reasoning.

Fellowship Academy

Advanced reconstruction, osteotomy, revision surgery, multi-ligament knee, cartilage restoration, patellofemoral surgery, arthroplasty nuance, research controversies and consultant-level judgment.

Consultant Masterclass

Expert debates, emerging technologies, complex decision-making, surgical pearls, failure analysis, literature updates and innovation.

Curriculum Matrix

Level	Depth	Primary Objective
Patient	Plain language	Understand symptoms, diagnosis and options
Medical Student	Foundation	Anatomy, pathology and clinical basics
MRCS	Applied core surgery	Examination, imaging and principles
FRCS	Specialist reasoning	Evidence, operations and complications
Fellowship	Advanced practice	Complex surgery and controversies
Consultant	Masterclass	Innovation, debates and expert judgement

5. Encyclopaedia Taxonomy

Anatomy

Bones, joints, cartilage, menisci, cruciate ligaments, collateral ligaments, posteromedial structures, posterolateral corner, anterolateral structures, extensor mechanism, tendons, bursae, synovium, neurovascular anatomy and surgical anatomy.

Biomechanics

Load transmission, contact pressure, kinematics, meniscal hoop stress, ligament restraint patterns, patellofemoral tracking, rotational stability, gait, sporting biomechanics and injury mechanisms.

Clinical Examination

History, inspection, palpation, range of motion, effusion, gait, ligament tests, meniscal tests, patellofemoral tests, paediatric assessment and complex knee examination.

Imaging

X-ray, ultrasound, MRI, CT, alignment imaging, advanced imaging, AI in imaging, weight-bearing imaging and Magic Angle MRI.

Conditions

Ligament injury, meniscal disorders, cartilage disorders, patellofemoral disorders, osteoarthritis, paediatric conditions, inflammatory disorders, tendon disorders, tumours and infection.

Surgery

Arthroscopy, meniscal repair and transplantation, ACL, PCL, MCL, PLC, ALL, osteotomy, cartilage restoration, MPFL, trochleoplasty, partial knee replacement, total knee replacement and revision surgery.

Rehabilitation

Principles, condition-specific rehabilitation, post-operative protocols, return to sport, return to work, strength, mobility, load management and recovery.

6. Website Style Aesthetic and User Experience

Design Aim

The website should combine the authority of a medical school, the usability of a learning platform and the visual appeal of a premium technology brand.

Colour Direction

Primary deep forest green, warm ivory background, restrained brushed gold accents, muted teal for learning highlights, soft grey for structure.

Homepage Principle

The homepage must be search-first. A user should instantly understand that this is where to learn anything about the knee.

Core Interactive Features

Large central search bar, learning pathway cards, interactive knee model, normal-versus-pathology imaging sliders, quiz widgets, case challenges and progress tracking.

Page Design

Short sections, collapsible advanced content, summary boxes, key learning points, images, diagrams, videos, self-assessment and related links. Avoid long unbroken textbook-style prose.

Mobile-First Requirement

Most users will access pages on mobile devices. Every page must be readable in short vertical sections, with clear headings, large images and minimal visual clutter.

7. Content Production SOP

Operating Principle

The assistant's role is knowledge management, source organisation, prompt execution, editorial review and evidence verification. The consultant's role is high-value clinical review.

Sequential Workflow

Understand site structure; select topic; gather sources; organise repository; upload to AI workspace; create content brief; generate draft; run editorial checks; verify factual claims; prepare consultant review pack; complete SEO; publish.

Topic Selection

Prioritise foundational and high-search topics first: ACL, meniscus, osteoarthritis, knee pain, MRI, anatomy, examination and rehabilitation.

Drafting Rule

AI output is a first draft only. It must never be published without structured quality control.

Version Control

Every article must have a draft version, checked version, consultant-reviewed version and published version.

Production Status Fields

Field	Purpose
Topic ID	Unique page number or taxonomy code
Priority	High, medium or low
Source status	Not started, partial, complete
Draft status	Not started, AI draft, edited
QA status	Assistant checked, evidence checked, consultant reviewed
Publication status	Ready, scheduled, published, refresh due

8. Assistant Training Manual

First Week Training

Read the sitemap, understand the learner levels, review the article template, learn the folder structure, practise source collection and produce three supervised sample drafts.

Core Responsibilities

Maintain source repository, track topics, upload sources, run prompts, format output, check references, highlight uncertainty, prepare review packs and update publication tracker.

Boundaries

The assistant should not independently change clinical conclusions, invent missing references, rewrite complex surgical nuance without review, or publish unreviewed material.

Daily Checklist

Open content tracker; confirm topic priority; gather sources; update repository metadata; create draft; check citations; log uncertainties; prepare review package.

Escalation Triggers

Conflicting evidence, unclear surgical recommendation, unsupported statistic, patient safety issue, commercial claim, drug or supplement claim, or any statement that could be construed as personal medical advice.

9. Source Acquisition SOP

Source Categories

Textbooks, review articles, landmark papers, guidelines, consensus statements, society resources, lecture slides, surgical technique notes, internal research, imaging examples and video assets.

Priority Sources

Own publications and presentations; BASK resources; ESSKA resources; ISAKOS resources; PubMed papers; Cochrane reviews; major textbook chapters; high-quality imaging teaching material.

Source Metadata

Every source must be logged with title, author, year, source type, topic, file location, relevance, quality level and notes.

Copyright Rule

Sources are for internal knowledge generation and review. Do not reproduce copyrighted text, figures or tables without permission. Use paraphrasing and original diagrams where possible.

Minimum Source Standard

Each substantial article should have at least one textbook or review source, several peer-reviewed papers where relevant and clear identification of any expert opinion.

10. AI Knowledge Base and Upload Workflow

Topic-Specific Uploads

Do not upload the entire library at once. Build focused knowledge bases such as ACL, Meniscus, Patellofemoral, Osteotomy or Arthroplasty.

Upload Pack

For each topic, upload the source summary, key papers, textbook excerpts where licensed, internal slides, relevant images and the article brief.

Context Discipline

The AI engine should be instructed to use only uploaded or specified sources for technical claims and to identify uncertainty rather than invent missing details.

Output Control

Request structured output aligned to KneeSchool templates. Ask for citations and source mapping.

Post-Generation

Export the draft into the article tracker folder and mark it as AI draft, not final copy.

11. AI Prompt Library

Prompt Philosophy

Prompts should specify role, topic, audience, source constraints, structure, tone, output format, citations, safety boundaries and commercial restrictions.

Universal Article Prompt

Use for standard encyclopaedia pages covering a topic across all learner levels.

Patient Page Prompt

Use for plain-language patient education. Avoid jargon, avoid diagnosis certainty, avoid direct personal medical advice and include when to seek clinical assessment.

Trainee Prompt

Use for MRCS and FRCS content. Emphasise applied anatomy, examination, imaging, differential diagnosis, management options, evidence and viva reasoning.

Fellowship Prompt

Use for advanced surgical topics. Include indications, contraindications, technique options, pitfalls, complications, failure analysis, rehabilitation and controversies.

Case Prompt

Use for imaging or clinical cases. Include presentation, images to insert, questions, model answer, teaching points and level.

Quiz Prompt

Use for MCQs, SBAs and viva questions with answer explanation and references.

SEO Prompt

Use after clinical approval to generate title, meta description, FAQ, schema suggestions and internal links.

Reusable Prompt Examples

Universal encyclopaedia page prompt:

You are an expert knee surgeon, anatomy teacher and medical education writer. Generate a KneeSchool page on [TOPIC] for Patient, Medical Student, MRCS, FRCS and Fellowship levels. Use UK English. Use only the provided sources for factual claims. Include overview, key learning points, anatomy, biomechanics, clinical relevance, imaging, management, rehabilitation, controversies, references, FAQs and related page suggestions. Identify uncertainty rather than inventing facts.

Patient page prompt:

Explain [TOPIC] to an educated patient. Use clear language, avoid unnecessary jargon, explain symptoms, diagnosis, treatment options, recovery, red flags and when to seek specialist assessment. Do not provide individual medical advice. Include FAQs and links to related KneeSchool pages.

FRCS viva prompt:

Create an FRCS viva preparation page on [TOPIC]. Include applied anatomy, classification, examination, imaging, management algorithm, operative options, complications, evidence controversies, common examiner questions and model answers.

12. Editorial Standards Manual

Language

Use UK English. Write in clear, direct prose. Avoid exaggerated claims. Avoid unsupported marketing language.

Medical Safety

Patient-facing pages must explain that information is educational and not a substitute for professional assessment.

Terminology

Use full terms on first use and abbreviations thereafter. Keep terminology consistent across the site.

Structure

Use predictable headings and short paragraphs. Each page should have overview, key learning points, main content, FAQs, references and related pages.

References

Do not allow fabricated references. Every named study, guideline or statistic must be checked.

Commercial Tone

Commercial links must be contextual and educational, not intrusive. Education must remain primary.

13. Quality Assurance Manual

QA Tiers

Tier 1 AI generation; Tier 2 assistant editorial review; Tier 3 source verification; Tier 4 consultant review; Tier 5 publication review.

Assistant QA

Checks structure, spelling, headings, references, internal links, duplication, clarity and formatting.

Evidence QA

Checks every factual claim that could affect clinical understanding. Statistics, surgical indications and treatment claims require source support.

Consultant QA

Focuses on accuracy, nuance, omissions, controversies, clinical judgement, surgical pearls and safety.

Publication QA

Checks formatting, SEO, links, images, alt text, disclaimers, metadata and page status.

Red Flags

Unsupported statistics, overconfident claims, non-evidence-based supplement claims, incorrect anatomy, wrong examination interpretation, outdated surgical recommendations or fabricated citations.

QA Checklist

- ☐ Correct page template used
- ☐ All learner levels present where required
- ☐ References checked
- ☐ No fabricated citations
- ☐ Clinical claims supported
- ☐ Commercial claims compliant
- ☐ Images anonymised
- ☐ SEO fields complete
- ☐ Consultant review completed

14. SEO and Internal Linking Manual

SEO Principle

Search visibility should be achieved through depth, clarity, topic authority and structured internal linking rather than keyword stuffing.

Topic Clusters

Build clusters around ACL, meniscus, arthritis, patellofemoral pain, knee replacement, MRI, rehabilitation and paediatric knee.

Each Page Requires

SEO title, meta description, slug, H1, H2 structure, FAQs, related articles, internal links and image alt text.

Internal Links

Every page should link upwards to its parent category, sideways to related topics and downwards to more advanced pages.

Commercial Links

Only use commercial links where clinically relevant. For example, an osteoarthritis page may link to consultation, imaging, rehabilitation and supplement education.

Content Refresh

High-value clinical pages should be reviewed every 6-12 months, or sooner if guidelines or major evidence changes.

15. Image and Video Standards

Visual Purpose

Images and video should clarify anatomy, pathology, imaging, surgical principles and rehabilitation. They should not be decorative filler.

Image Categories

Anatomy diagrams, MRI examples, X-ray examples, ultrasound images, arthroscopy images, rehabilitation photographs, surgical illustrations and infographics.

Video Categories

Patient explanations, clinical examination, imaging interpretation, surgical technique, rehabilitation exercise, expert interviews and conference teaching.

Image Rights

Use original images where possible. For external figures, obtain permission or create original redrawn educational diagrams.

Alt Text

Every image requires descriptive alt text for accessibility and SEO.

Quality Control

All clinical images must be anonymised and checked before upload.

16. Sportshealing Integration Strategy

Principle

Sportshealing should be presented as the clinical pathway for users who need expert assessment, imaging, diagnosis, injections, surgery or rehabilitation.

Appropriate Placement

Condition pages, imaging pages, treatment pages, rehabilitation pages and complex case pages.

Call-To-Action Examples

Need a specialist knee assessment? Learn about Sportshealing consultations. Need imaging? Learn about MRI and ultrasound pathways. Considering surgery? Learn about specialist knee surgical assessment.

Avoid

Aggressive sales language, excessive pop-ups, unsupported claims or implying all users need private care.

Conversion Assets

Consultation pages, imaging pathway, knee score test, patient questionnaires, outcome measures and educational downloads.

17. OmKneeHealth Integration Strategy

Principle

OmKneeHealth should be integrated as a supportive musculoskeletal health and recovery product pathway, not as a replacement for diagnosis or treatment.

Appropriate Placement

Nutrition, collagen, vitamin D, osteoarthritis, recovery, strength, women's knee health, ageing knee and prevention pages.

Compliance Standard

Avoid disease treatment claims unless supported and legally compliant. Use cautious educational language.

Product Education

Explain ingredients, rationale, evidence level, cautions and who should seek medical advice before use.

Commercial Format

Use subtle product cards, related resources and education-led links rather than banner advertising.

18. Technology Stack and AI Tools

Website Platform

Choose a scalable CMS or hybrid stack that supports structured content, search, taxonomy, membership, video, quizzes, SEO and future AI functionality.

Core Requirements

Fast search, content tagging, user levels, reusable templates, quiz functionality, media library, SEO metadata, analytics and secure admin roles.

AI Workflow Tools

A document-aware AI engine, citation-aware outputs, prompt templates, topic knowledge bases, version control and human review workflow.

Content Management

Use a master tracker in spreadsheet or project management software with status fields for source collection, drafting, QA, consultant review, SEO and publication.

Future AI Tutor

Ask KneeSchool should answer questions at the user's selected educational level, with citations and safe boundaries.

19. KPIs and Reporting

Production KPIs

Sources collected, topics briefed, drafts generated, drafts checked, consultant-reviewed pages, published pages and pages refreshed.

Quality KPIs

Reference accuracy, number of flagged uncertainties, consultant revision burden, factual error rate and publication readiness.

SEO KPIs

Indexed pages, impressions, clicks, ranking keywords, topic cluster growth, internal link coverage and organic traffic.

Engagement KPIs

Search usage, time on page, quiz completion, video views, return users, newsletter sign-ups and learning pathway progress.

Commercial KPIs

Sportshealing enquiries, knee score completions, imaging enquiries, consultation bookings, OmKneeHealth click-throughs and product conversions.

Reporting Frequency

Weekly operational update, monthly content report, quarterly strategy review.

20. 18-Month Delivery Roadmap

Months 1-3 Foundation

Complete sitemap, brand direction, templates, source repository, article tracker, assistant workflow and first 100 pages.

Months 4-6 Core Encyclopaedia

Build anatomy, biomechanics, examination, imaging and high-search conditions. Target 250 total pages.

Months 7-9 Conditions Library

Expand ligament, meniscus, cartilage, patellofemoral, arthritis, paediatric and women's knee health sections.

Months 10-12 Surgery Academy

Build procedure pages, surgical decision-making, complications, rehabilitation and advanced trainee content. Target 600 pages.

Months 13-15 Research Academy

Build landmark paper summaries, journal club, methodology, controversies and evidence updates.

Months 16-18 Video and Advanced Features

Launch case library, video academy, quizzes, AI tutor beta, membership or course structure. Target 1000+ pages.

Roadmap Summary Table

Phase	Months	Primary Deliverable	Target
Foundation	1-3	Architecture, templates, repository	100 pages
Core Build	4-6	Anatomy, biomechanics, imaging	250 pages
Conditions	7-9	Major pathology libraries	400 pages
Surgery	10-12	Surgery academy	600 pages
Research	13-15	Landmark papers and controversy hub	800 pages
Advanced	16-18	Video, cases, AI tutor	1000+ pages

21. Appendices and Templates

Appendix A

Master article tracker template.

Appendix B

Source metadata sheet.

Appendix C

Article brief template.

Appendix D

QA checklist.

Appendix E

Consultant review form.

Appendix F

SEO checklist.

Appendix G

Publication checklist.

Appendix H

Prompt library examples.

Appendix I

Weekly report template.

Appendix J

Risk register.

Appendix A - Master Article Tracker Template

Topic ID	Topic	Category	Level	Priority	Sources gathered	AI draft	Assistant QA	Evidence QA	Consultant review	SEO complete	Published	Refresh date

Appendix B - Source Metadata Template

Source ID	Topic	Title	Author	Year	Source type	Journal/ book/ society	File path	Quality rating	Key points	Usage notes

Appendix C - Article Brief Template

Topic	Audience levels	Purpose	Required sections	Key sources	Must include	Must avoid	Commercial relevance	Images required	Reviewer notes

Appendix D - Consultant Review Form

Article	Clinical accuracy	Missing concepts	Incorrect statements	Evidence concerns	Surgical nuance	Commercial appropriateness	Approved	Required changes

Appendix E - SEO Checklist

SEO title	Meta description	Slug	H1	H2s	FAQs	Internal links	Alt text	Schema	Related articles	CTA

Appendix F - Weekly Report Template

Week commencing	Sources added	Drafts generated	Pages checked	Pages approved	Pages published	Risks	Questions for consultant	Next week priorities

Appendix G - Risk Register Template

Risk	Likelihood	Impact	Owner	Mitigation	Status